

AI/ML Capstone Project Proposal

Research Question

Can large language models (LLMs) accurately classify non-executable files (e.g., scripts) as malicious or benign using minimal metadata (such as headers and footers) without relying on traditional signature-based detection?

Expected Data Sources

- Datasets will be from Microsoft Defender sample store of known malicious and good scripts. Malicious and clean labels will come from the Defender ML team's internal standard file labels for ML models.
 - Focus will be on initial script files at initial access stage of attacks at web download (malvertising, social engineering, etc).
 - Clean file samples sourced from open-source projects and public code repositories (e.g., GitHub).
 - Synthetic datasets generated for controlled experiments.
-

Techniques for Analysis

- **Data Preprocessing:** Extract header/footer and structural features from non-PE files.
 - **LLM-based Classification:** Fine-tune or prompt large language models for binary classification (malicious vs. benign).
 - **Evaluation Metrics:** Precision, recall, F1-score, and ROC-AUC to measure detection performance.
 - **Optional Extensions:** Use LangGraph for workflow orchestration and implement a script labeling system for future ML training.
-

Expected Results

- Demonstrate that LLMs can achieve meaningful recall at a high precision in detecting malicious non-executable files using lightweight metadata.
Label unlabeled script files to improve the classical ML models that we deploy at large scale. The lack of labels on script files is a current notable limitation of our existing deployed protection models.
 - Highlight gaps where classical ML models fail and propose improvements.
-

Why This Question Is Important

This research is driven by curiosity about how advanced AI techniques can tackle emerging cybersecurity challenges. The focus is on the importance of protection against initial access to script files that often lead to ransomware attacks.

Non-executable files, such as scripts, are an interesting category because they often lack clear labeling and are harder to classify using traditional methods. Using FakeUpdates as an example, our classical ML models are struggling due to lack of file label quality for script files. Exploring whether LLMs can improve detection in this space provides an opportunity to:

- Experiment with novel approaches to file classification.
- Contribute to academic and industry understanding of AI applications in security.
- Generate insights that could inspire future research, without assuming any current shortcomings in existing solutions.